

TABLE 1
USE CASES, FEATURES AND FUTURE TRENDS OF HYPER-REAL PLATFORMS

Use Case	Key Features	Future Trends
Gaming and entertainment	Highly realistic graphics and physics simulation	Integration of haptic feedback for enhanced immersion
	Immersive audio and spatial sound	Seamless integration of virtual and augmented reality
	Interactive gameplay and lifelike characters	Real-time multiplayer experiences across platforms
Architecture and design	Detailed material simulations	Integration of AI for automated design optimisations
	Accurate lighting effects	Real-time collaboration and feedback during design process
	Virtual walkthroughs and spatial interactions	Integration with Building Information Modelling (BIM)
Education and training	Realistic simulations of various scenarios	Adaptive learning experiences based on user interactions
	Hands-on training in safe environments	Integration of natural language processing for dialogue
	Experiential and immersive learning	Virtual mentorship and guidance for skill development
Retail and e-Commerce	Immersive 3D virtual stores	Personalised recommendations based on user preferences
	Accurate product visualisation	Integration of virtual try-on technologies
	Interactive product interaction	Integration with blockchain for secure transactions
Healthcare and telemedicine	Virtual consultations in realistic environments	Real-time patient monitoring and vital sign integration
	Medical simulations for training and practice	Integration with medical imaging technologies
	Enhanced visualisation for accurate diagnosis	Integration of telepresence technologies for remote surgery

ray tracing and advanced rendering techniques allow hyper-real platforms to achieve stunning visual fidelity.

High-resolution textures, realistic lighting, and lifelike animations contribute to a more authentic experience. AI algorithms enable dynamic and intelligent interactions within hyper-real platforms. Machine learning techniques improve user interactions, object recognition, and natural language processing, enhancing overall realism. Hyper-real platforms revolutionise gaming experiences, providing realistic simulations, immersive storytelling, and interactive gameplay. Users can engage in virtual sports, explore fantastical worlds, and connect with other players in hyper-realistic environments. Hyper-real platforms offer enhanced virtual conferencing capabilities, enabling lifelike interactions, spatial audio, and realistic environments.

Remote collaboration becomes more immersive, fostering better communication and engagement

among participants. Industries such as aviation, healthcare, and defence utilise hyper-real platforms for training simulations. Realistic scenarios enable trainees to practice skills, make critical decisions, and experience lifelike consequences in a safe environment.

Architects and designers leverage hyper-real platforms to create virtual walkthroughs and visualisations of buildings and spaces. Clients can experience and provide feedback on architectural projects before they are physically constructed. Hyper-real platforms facilitate virtual therapy sessions, pain management, and phobia treatment through immersive experiences.

Patients can engage in simulations that expose them to controlled environments, enabling gradual exposure and therapeutic interventions. Hyper-real platforms raise concerns about data privacy, consent, and potential misuse of realistic virtual scenarios.

Ensuring user safety, consent, and responsible use of personal

information is crucial. Technical limitations and infrastructure requirements play a vital role in hyper-real platform design. Achieving hyper-realistic experiences requires powerful hardware and infrastructure, which may limit accessibility for some users. Bandwidth requirements, latency issues, and device compatibility pose challenges for widespread adoption. Creating hyper-realistic content requires significant resources, expertise, and time. User-generated content platforms and tools need to be developed to empower users to contribute to the metaverse.

Hyper-real platforms should prioritise accessibility features to cater to users with disabilities. Ensuring inclusivity and designing interfaces that accommodate diverse user needs is essential. These platforms have the potential to redefine social interactions, allowing people to connect and communicate in new ways. Relationships may be formed and nurtured within virtual environments, challenging traditional